

Understanding Mobile Information Needs and Behaviours

Yonchanok Khaokaew
y.khaokaew@unsw.edu.au
PhD Student
School of Computer Science and
Engineering
University of New South Wales
NSW, Australia

Flora D. Salim
flora.salim@unsw.edu.au
PhD Supervisor
School of Computer Science and
Engineering
University of New South Wales
NSW, Australia

Hao Xue
hao.xue1@unsw.edu.au
PhD Co-Supervisor
School of Computer Science and
Engineering
University of New South Wales
NSW, Australia

ABSTRACT

Smartphones have become an integral part of our daily lives, with an ever-increasing number of users relying on mobile applications to meet their needs. This widespread usage necessitates a deep understanding of how people utilize their devices to develop more effective features. Leveraging context-based information from these devices offers insights into user needs and behaviours. This PhD study focuses on three primary aspects: deciphering context signals to determine user needs, modelling mobile user behaviour based on these signals, and generating synthetic smartphone user patterns. Our goal is to provide invaluable data that can drive the evolution of future mobile applications, further enhancing the user experience.

CCS CONCEPTS

• **Information systems** → **Data mining; Spatial-temporal systems; • Human-centered computing** → **Empirical studies in ubiquitous and mobile computing; • Computing methodologies** → **Machine learning; Model development and analysis.**

KEYWORDS

Mobile User Behaviour Modelling; App Usage Prediction; Large Language Models; Synthetic Data; Clustering

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1 INTRODUCTION

In today's world, smartphones have transitioned from luxury to necessity, transforming into an essential tool that offers a myriad of functionalities accessible at any time and place. Their ubiquitous presence shapes our experiences, as the numerous applications at our fingertips empower us to fulfil various needs. Mobile applications form the backbone of modern digital interactions; hence,

understanding their usage patterns can be a significant step in understanding the users themselves.

Before diving further, it's pertinent to define what we mean by *context* in the scope of this research. Context, as perceived in this study, encompasses the passive information that can be retrieved from smartphones. This includes but isn't limited to, current location translated to points of interest, time of the day, the number of meetings scheduled in the user's calendar, physical activity like steps or movement, and historical patterns of app usage. Each of these data points provides insights into the user's environment, preferences, and behaviour, making them essential tools in predicting and tailoring the user's smartphone experience.

With the rise of smartphones as intelligent tools that can comprehend our context, interpret our needs, and offer fitting solutions, there's been an upsurge in smartphone usage over the past decade. This has made it imperative to deepen our understanding of users' needs and behaviours to better tailor future applications and features.

Our research seeks to elucidate the patterns of current smartphone usage and strategise how best to address users' needs through mobile applications. This investigation is segmented into three pivotal components: deciphering user needs through the context signals, crafting a predictive model based on this contextual data, and generating synthetic smartphone user behaviours. The insights from this endeavour aim to influence the design and development of future mobile applications, enhancing user experience and cementing the smartphone's indispensable role in daily life.

In the first phase, our attention is on the needs of smartphone users, particularly regarding their interaction with digital assistants (DAs). While initially designed as task facilitators, these DAs have emerged as pivotal for understanding and predicting user behaviour. By leveraging the context as defined, we aim to dissect the relationship between a user's immediate environment and their consequent needs. This understanding paves the way for more personalized user experiences, providing adaptability that resonates with individual preferences.

The second phase focuses on modelling user behaviours derived from the defined context. Central to this is the prediction of app usage. Given apps' predominant role in smartphone engagement, predicting their use becomes paramount for enhancing user interaction. By anticipating user actions, devices can pre-load or pre-fetch parts of prospective apps, ensuring swifter launch times and diminished delays. This is particularly invaluable for high-resource apps. Our designs are built around individual habits and preferences, keeping the device attuned to the user's needs. However, data collection poses challenges, especially around privacy concerns and

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resource limitations. We're pioneering a synthetic data generation technique that emulates user behaviour across diverse datasets to counter this. This synthetic data has potential applications in multifaceted tasks, from app usage prediction to foreseeing location trajectories. This methodology safeguards participant privacy and offers a rich palette of behavioural insights. Through these investigations, we aspire to deepen comprehension of mobile user behaviours and elevate the overall user experience.

2 RELATED WORK

This section provides a comprehensive overview of the existing body of research relevant to understanding mobile information needs and behaviour, focusing on three principal areas: mobile information needs, app usage prediction, and synthetic data generation.

2.1 Mobile Information Needs

The rapid progression and omnipresence of mobile technology have significantly impacted how users access information, leading to a shift in information needs. The research by Sohn et al. [18] indicated that mobile information access behaviour tends to be more context-dependent than traditional methods. A similar perspective was shared by Church et al. [6], suggesting that evolving contexts, including location, time, activity, and social circumstances, significantly impact information needs.

In a later study, Church et al. [5] pointed out an increasing trend of users accessing mobile web services from non-mobile locations, such as their homes or workplaces, representing a substantial shift from their prior work [6]. Despite this shift, the importance of user context in many mobile information scenarios was emphasized, underlining the need for more research into how the changing contexts influence mobile users' information needs.

2.2 Mobile App Usage Behaviour

Various models and approaches have been proposed in the context of predicting app usage. Huang et al. [9] utilized Bayesian methods and a combination of features to predict the next app a user might use, achieving a noteworthy accuracy level. Shin et al. [17] offered a personal next-app prediction model that capitalized on an extensive range of smartphone contextual features with significant accuracy. Other researchers, like Liao et al. [14], introduced feature selection to choose useful features for each user, subsequently using a kNN classifier for prediction. Xu et al. [21] explored the potential of social context information to enhance the accuracy of app usage prediction models. Later, Parate et al. [15] proposed that models could rely solely on the history of app usage sequence to deliver accurate predictions, eliminating the need for privacy-sensitive context features. Similarly, Srinivasan et al. [19] leveraged smartphones' context data to devise a method of mining user behaviour patterns directly on devices. This technique accentuates user privacy while diminishing cloud dependencies. Their analysis of 106 participants underscored the method's efficiency in pinpointing co-occurrence patterns, providing pivotal insights for enhancing UI and app functionalities.

More recent research efforts include the development of AppUsage2vec by Zhao et al. [24], which incorporated an app-attention

mechanism and a dual deep neural network model to improve prediction accuracy. Recent work by Kang et al. [10] attempts to predict app usage in specific contexts (commuting). In the study, a set of contextual information is presented that is essential for predicting the following used apps, considering users are moving. Their proposed model can provide high accuracy in app usage prediction while moving based on this set of contextual information.

These previous studies have demonstrated the effectiveness of using contextual information to enhance the modelling performance of the app usage prediction model. The proposed models can solve the app usage prediction problem using a specific type of contextual feature and provide high performance. In some situations, however, these types of contextual features may not be available. Therefore, we plan to develop a model that can be applied to different types of contextual information in the second part of my research.

2.3 Synthetic Data Generation

Synthetic data generation represents an important avenue for protecting sensitive information and expanding the available labelled data for training machine learning models [3]. Many different methods have been used to generate synthetic data [1, 7, 8, 25]. Alkurd et al. [1] proposed a synthetic data generation model for predicting user satisfaction in real-time. The Generative Adversarial Networks (GANs) approach has been employed by researchers like Delaney et al. [8], and Zheng et al. [25] for creating synthetic ECG signals and phasor measurement unit (PMU) data, respectively.

Recently, Yoon et al. [22] introduced a model capable of producing high-fidelity and privacy-preserving synthetic data. However, a common challenge among these studies is the focus on specific domains, while the generation of synthetic data for mobile information needs and behaviour remains underexplored.

This existing body of work provides valuable insights into understanding mobile information needs and behaviours but also illuminates the need for additional research, particularly in generating synthetic data for mobile information needs and behaviour modelling. This project aims to fill these gaps in our current understanding and contribute meaningfully to the field.

3 METHODOLOGY

Our methodology aims to address three goals: *understanding context features*, *modelling mobile user behaviour*, and *addressing mobile user behaviour while assuring privacy*. Three comprehensive components characterize the approach:

3.1 Contextual Feature Analysis

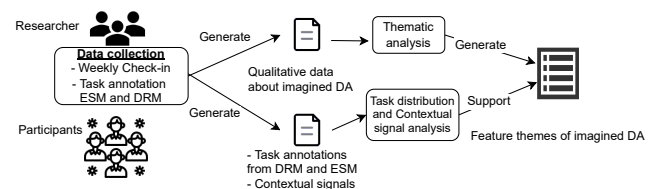


Figure 1: Overview of the approach for contextual feature analysis

The primary aim of our first component is to analyze user tasks and their context meticulously. We collect a contextually rich dataset through the *Experience Sampling Method (ESM)* and the *Daily Reconstruction Method (DRM)*, offering an extensive understanding of daily tasks across various occupations. The ESM/DRM method was chosen over speculative design activities given their greater ecological validity [20]. ESM, which provides a comprehensive view of an individual's daily life, can provide insight into how tasks develop over time. The DRM method is designed to capture a full picture of a day, including the duration of daily activities, without disrupting daily life [2]. DRM can provide complementary information about a person's activities daily. These methods collectively allow us to discern what type of contextual support users might need when undertaking tasks, informing our research's subsequent stages.

Our analysis of the collected data unfolds in two stages, collectively guiding our understanding of contextual feature importance. In the first stage, we focus on the qualitative interview data collected weekly, using thematic analysis to uncover workers' needs for imagined Digital Assistants (DAs), particularly in the realm of time, task, and information management. This analysis provides an initial insight into the context-dependent nature of these needs, offering a qualitative perspective on important contextual features. The second stage dives into the quantitative realm. We use the sensing data collected from participants to confirm the demand for these identified needs, validating and quantifying the significance of the identified contextual features. This stage involves categorizing participants into clusters using contextual signals, thereby establishing a direct relationship between the context and the users' behaviour or needs. Based on the characteristics of each cluster, we deduce the potential demand within each group. This inferred demand then contrasts with the needs identified from the previous thematic analysis. In doing so, we can quantify the significance of each contextual feature, drawing an empirical link between context and user needs. Together, these two stages allow us to understand the contextual features and their relevance in different scenarios and fulfil our *Contextual Feature Analysis* goal. Further details about this methodology are documented in our previous work [11].

3.2 Mobile User Behaviour Modeling with LLMs

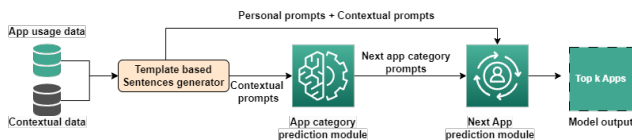


Figure 2: Overview of the proposed model for modelling mobile user behaviour with LLMs

The second component is pivotal as it aims to model mobile user behaviour using *Large language models (LLMs)*. Our decision to utilize LLMs is grounded in its ability to effectively handle the heterogeneity of the contextual information and effectively capture the user's context. The overview of the proposed model is illustrated in Figure 2. Mobile applications form the backbone of modern digital interactions; hence, understanding their usage patterns can be a significant step in understanding the users themselves. Our method

allows us to accurately predict mobile app usage leveraging a robust language model to address the heterogeneity of the contextual information problem.

Our model unfolds in three primary segments. Firstly, the *Template-based Contextual Sentence Generation Module* considers app and context data, generating sentences that encapsulate the user's current mobile usage scenario. Rich in context, these sentences spring from templates that effortlessly integrate relevant details such as the time, location, and app categorization. In the second phase, dubbed the *App Type Prediction Training Stage*, our model is trained to predict the likely subsequent app category. It utilizes sentences crafted by the Template-based Contextual Sentence Generation Module and blends them with historical app data, leveraging the power of LLMs to predict the app type that will be used next. The final segment of our model, the *Next App Prediction Training Stage*, refines the model developed in the previous stage to predict imminent app usage. It achieves this refinement by merging the sentences used in the previous phase with those generated from the user's personal app usage history and the ensemble of installed apps. The final model, a product of this last stage, formulates an answer sentence indicating the next app likely to be used.

In sum, the proposed model harnesses the prowess of language models to adeptly handle contextual data heterogeneity and be able to model mobile user behaviour on the available contextual information.

3.3 Synthetic Data Generation

Privacy is the cornerstone of any user study. The final component of our methodology addresses this critical concern. In this subsection, we focus on our approach to generating synthetic data, leveraging a novel combination of *OpenAI's CLIP architecture* [16] and *variational autoencoder (VAEs)* [13] methodologies (see Fig. 3).

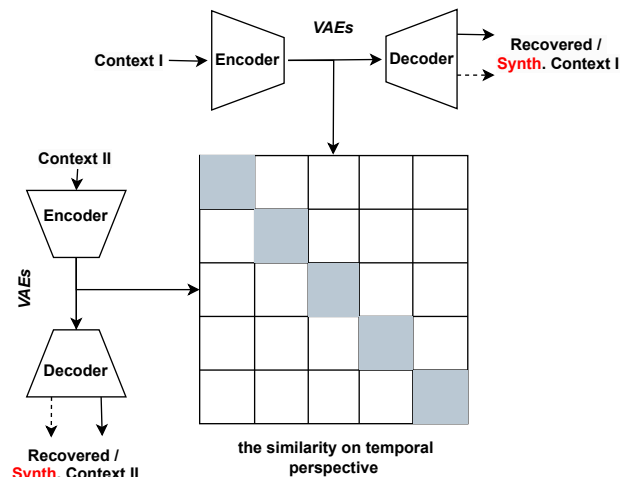


Figure 3: Overview of the proposed model for synthetic data generation part

The CLIP model has been recognized for its capability to understand and align multimodal data, such as images and text, using

a contrastive objective in the latent space. The essence of CLIP's success lies in its ability to align semantically similar information across distinct modalities. In our methodology, we utilize this model to align the encoded features of our proposed system, thereby ensuring that features from the same temporal (time) context are brought closer in the latent space. This process aids in learning robust and meaningful representations for each contextual feature, effectively capturing their relationships and variances.

Subsequently, the variational autoencoder, a class of generative models, is employed to synthesize new data. The autoencoder is split into two stages: an encoder that compresses the input data into a latent space representation and a decoder that reconstructs the input from this latent space representation. The strength of a VAEs lies in its probabilistic graphical modelling toolbox, which allows it to model complex, high-dimensional datasets using latent variables. In our case, the decoder part of the VAEs is used to generate the synthetic data. It takes the encoded data and reconstructs the original input. However, this 'original input' is, in fact, a synthesized version of the real data, ensuring the preservation of privacy while retaining the structural patterns of the original dataset. Thus, this dual-stage process harnesses the strength of both techniques: CLIP for its superior data alignment capabilities and variational autoencoder for its robust data synthesis properties.

In sum, our synthetic data generation strategy combines the prowess of the CLIP architecture in handling complex, multi-contextual data and the generative capabilities of an VAEs. This novel method ensures not only the creation of accurate, valuable synthetic data but also safeguards the privacy of user information. Further subsections will elaborate more intricately on each stage of the model, shedding light on our intricate approach.

4 EVALUATION

Our methodology encompasses a three-pronged approach aimed at achieving a comprehensive understanding of user behaviour, modelling this behaviour, and ensuring privacy throughout.

The first phase, Contextual Feature Analysis, is centred around compiling and analysing self-reported data alongside sensor data. This two-stage process commences with a qualitative assessment of user data, which is obtained via interviews and questionnaires. The primary goal during this stage is to uncover user preferences and expectations. In the subsequent stage, we execute a quantitative assessment using sensor data that we've collected to verify or dispute the user needs to be identified in the first stage. We will then analyze the individual clusters' characteristics to determine each group's potential demands based on their respective attributes. Following this, we compare these demands with the needs of the participants that were inferred from the thematic analysis carried out in the first stage. This comparison establishes that contextual information can serve as a reliable indicator of user needs. This process underscores our goal to validate the effectiveness of using context as a key informant in understanding user behaviour and demands.

Informed by insights from the first phase, the second phase, Mobile User Behaviour Modeling with LLMs, will apply the proposed model to predict app usage based on user behaviour patterns. The efficacy of this model will be validated through metrics such as

Top-1 and Top-k accuracy, as well as the Mean Reciprocal Rank (MRR) score of prediction for the subsequent app to be utilized. To address the user cold start issue, User-Based Cross-Validation is performed, establishing a stringent evaluation benchmark. The proposed model will be assessed on public app usage datasets such as the China Telecom dataset used in previous works [4, 12, 23].

In the Synthetic Data Generation phase, the utility and privacy of the synthetic data will be evaluated through a multi-pronged approach. This begins with a statistical comparison between synthetic and real data, then proceeds to assess the usefulness of synthetic data through a performance comparison of models trained on synthetic and private data. Additionally, the Pointwise Differential Training Privacy (PDTP) metric will be applied to ensure privacy. We will also compare the distribution of reconstruction errors and investigate the potential for de-identification in the synthetic data, ensuring it can be used safely for future studies.

These components collectively contribute to a robust and comprehensive methodology, offering deep insights into user behaviour, effective behavioural modelling, and utmost respect for user privacy. We ensure computational efficiency across all methods and perform comparative analyses with existing methodologies to emphasize our approach's distinct advantages. The end goal is a comprehensive, reliable, and effective solution tailored to understanding user behaviour, modelling, and preservation of privacy.

5 EXPECTED CONTRIBUTION

Our research significantly contributes to understanding, modelling, and ethically handling mobile user behaviour.

Starting with the Contextual Feature Analysis, we extract valuable insights from self-reported and sensor data, enriching our understanding of user behaviour within unique contexts. In the Mobile User Behaviour Modeling with LLMs, we revolutionize the predictive modelling of app usage by harnessing the power of LLMs. This approach enhances prediction accuracy and marks a significant leap forward in user-centred AI systems. The results from this portion of our research are currently under review.

Respecting privacy concerns, our Synthetic Data Generation phase introduces an innovative approach to generating synthetic data that retains real data patterns while protecting privacy. This step could dramatically reshape mobile user behaviour studies, offering privacy-secure data exploration. To validate the Synthetic Data Generation, we establish comprehensive evaluation metrics, measuring the synthetic-real data difference, assessing synthetic data's practical usefulness, and comparing generated samples.

In summary, our research brings critical advancements to AI and mobile user behaviour studies by deepening user understanding, improving behaviour prediction modelling, pioneering a synthetic data generation method, and setting up robust evaluation metrics, all while prioritising user privacy.

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